

Math 162
Final Exam
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1. Determine whether each of the following functions is integrable on $[0, 1]$.

(a)

$$f(x) = \begin{cases} \frac{\sin x}{x} & \text{for } x \neq 0, \\ 1 & \text{for } x = 0. \end{cases}$$

(b)

$$f(x) = \begin{cases} \frac{\cos x}{x} & \text{for } x \neq 0, \\ 1 & \text{for } x = 0. \end{cases}$$

(c)

$$f(x) = \begin{cases} \frac{\log x}{x} & \text{for } x > 0, \\ 1 & \text{for } x = 0. \end{cases}$$

2. Prove that if $f, g : \mathbb{R} \rightarrow \mathbb{R}$ are differentiable and $h : \mathbb{R} \rightarrow \mathbb{R}$ is continuous, then

$$\frac{d}{dx} \int_{f(x)}^{g(x)} h(t) dt = h(g(x))g'(x) - h(f(x))f'(x).$$

3. (a) If $\sum_{n=1}^{\infty} a_n$ converges absolutely, prove that $\sum_{i=1}^{\infty} a_{n_i}$ converges for any subsequence $a_{n_1}, a_{n_2}, \dots$.

- (b) Give an example to show that this is false if we only assume that $\sum_{n=1}^{\infty} a_n$ converges.

4. Define $f : \mathbb{R} \rightarrow \mathbb{R}$ by

$$f(x) := \begin{cases} \frac{\cos(3x^5) - 1}{x^7} & \text{for } x \neq 0, \\ 0 & \text{for } x = 0. \end{cases}$$

Compute $f^{(2023)}(0)$. (Hint: Use Taylor series.)

5. (Extra credit) Prove that

$$1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots = \frac{\pi}{4}.$$

(Hint: Note that $\arctan(1) = \frac{\pi}{4}$.)